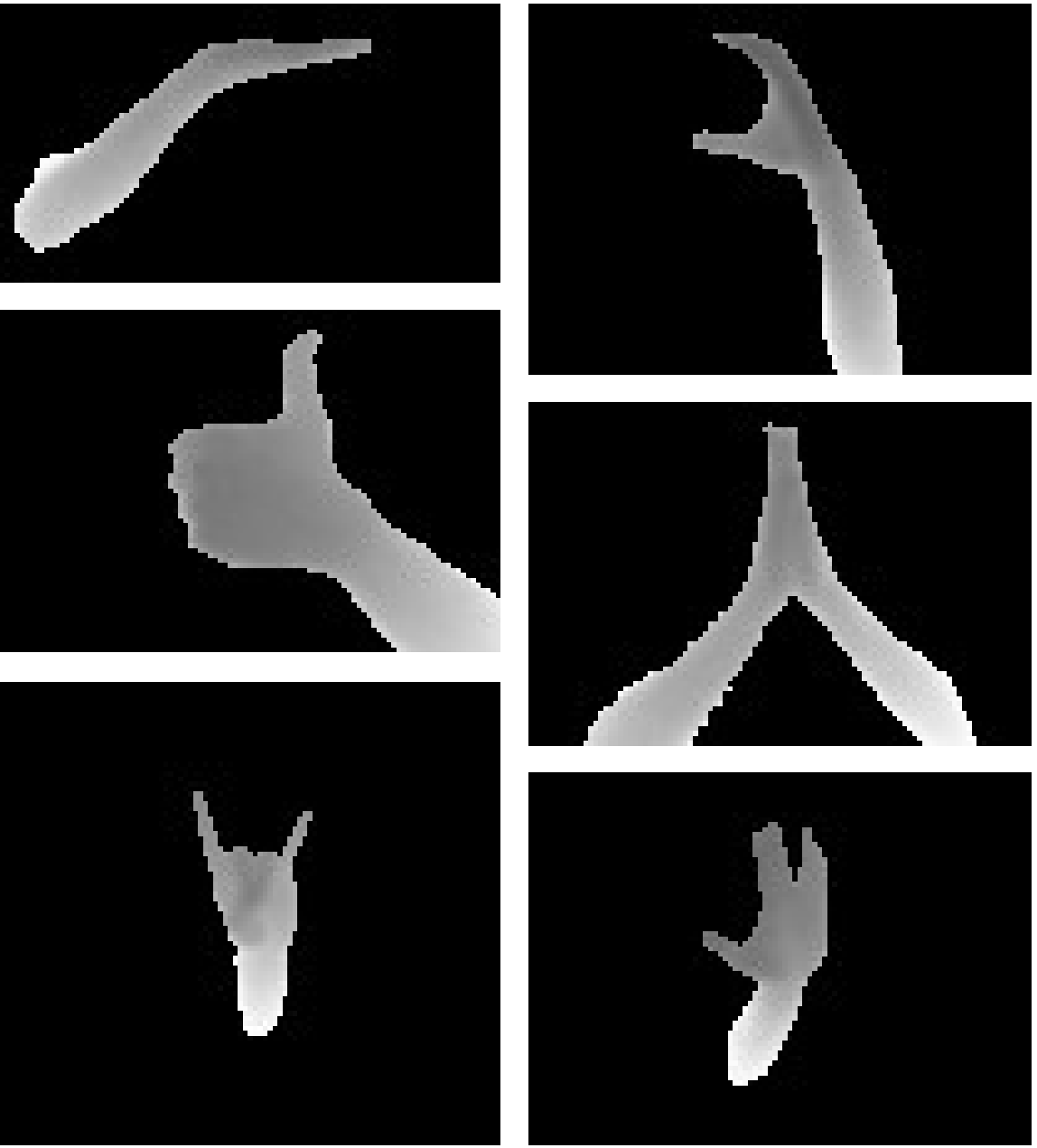

Projekt 2
Hetnerovič, Kocúr



Rozpoznávanie gest ruky z hĺbkovej mapy

Spracovanie dát

- Zmena veľkosti na 101 x 101
- Vloženie do np.array
- One-hot encoding
- Rozdelenie na train 64%, test 20%, val 16%

Vytvorenie CNN siete

- 4 konvolučné + pooling
- 4 plne prepojené

```
def create_custom_serial_model(number_of_classes):
    model = Sequential()

    model.add(Conv2D(32, kernel_size=(3, 3), padding='same', input_shape=(101, 101, 1)))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))

    model.add(Conv2D(64, kernel_size=(3, 3), padding='same'))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))

    model.add(Conv2D(128, kernel_size=(3, 3), padding='same'))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))

    model.add(Conv2D(256, kernel_size=(3, 3), padding='same'))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))
    model.add(Flatten())

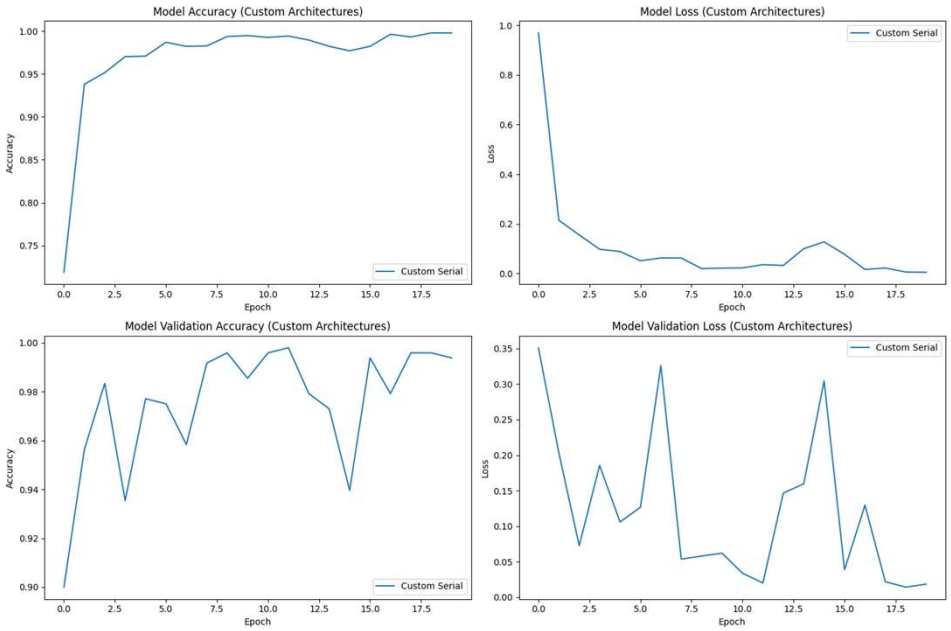
    model.add(Dense(1024))
    model.add(ReLU())
    model.add(Dense(256))
    model.add(ReLU())
    model.add(Dense(128))
    model.add(ReLU())
    model.add(Dense(64))
    model.add(ReLU())
    model.add(Dropout(0.5))

    model.add(Dense(number_of_classes, activation='softmax'))
    model.compile(optimizer=Adam(), loss='categorical_crossentropy', metrics=['accuracy'])

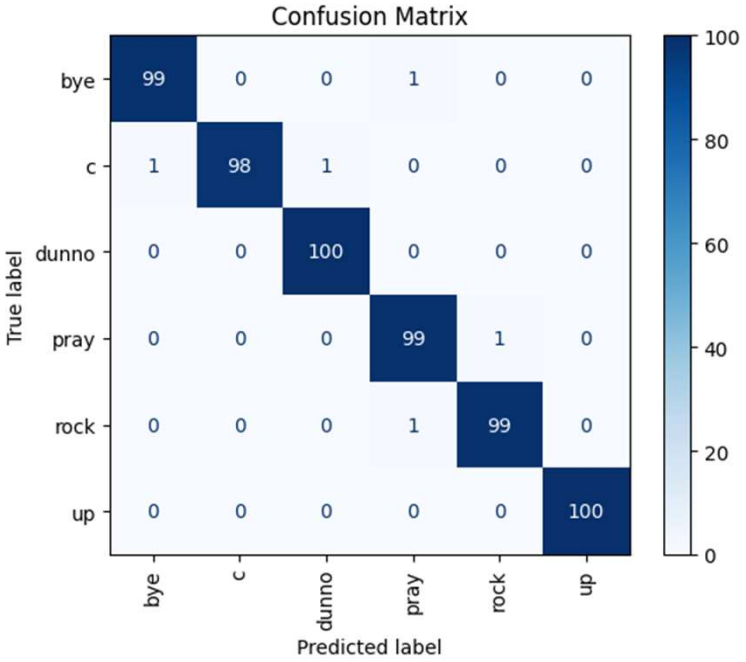
    return model
```

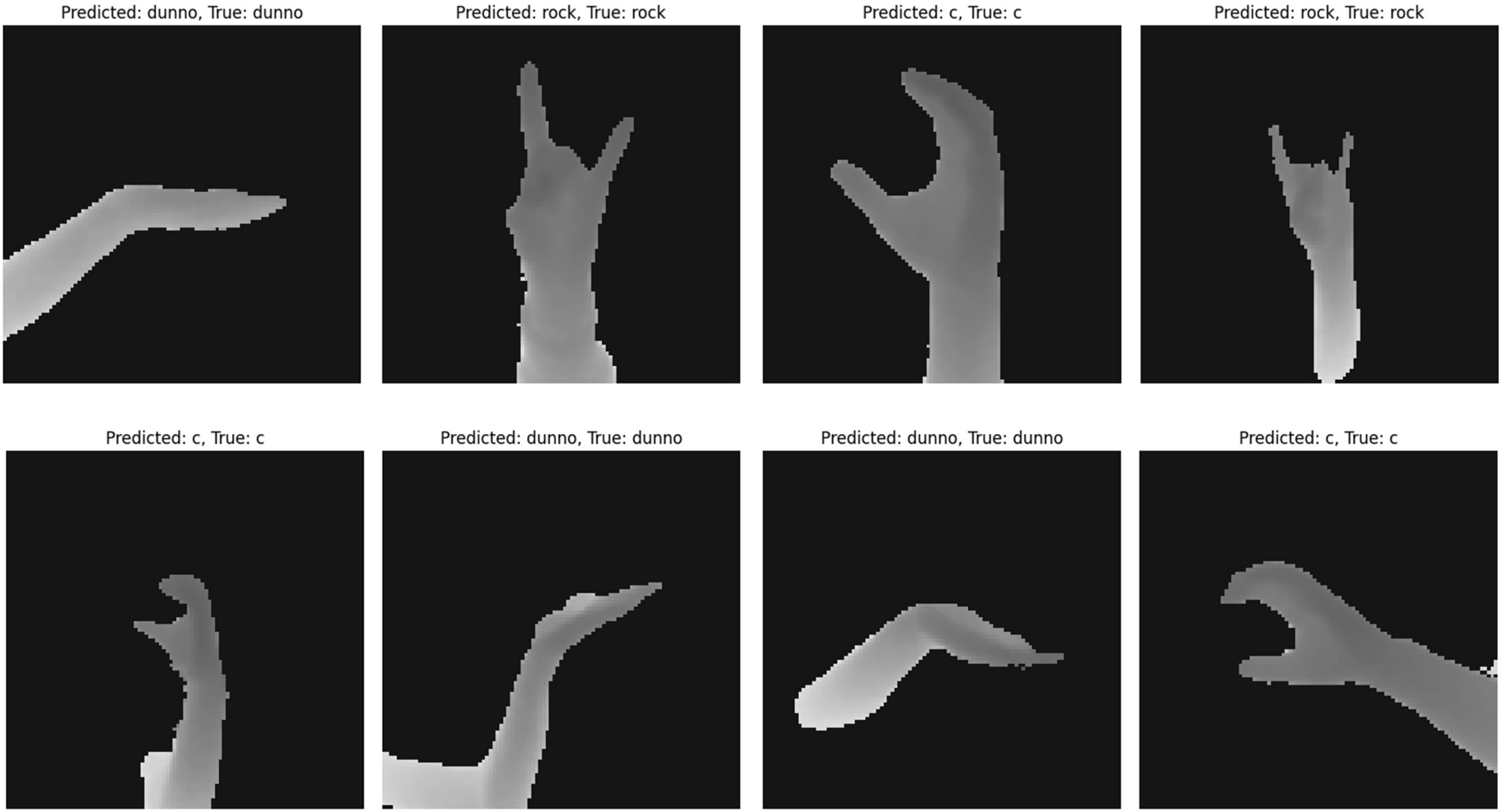
Výsledky trénovania

Proces trénovania

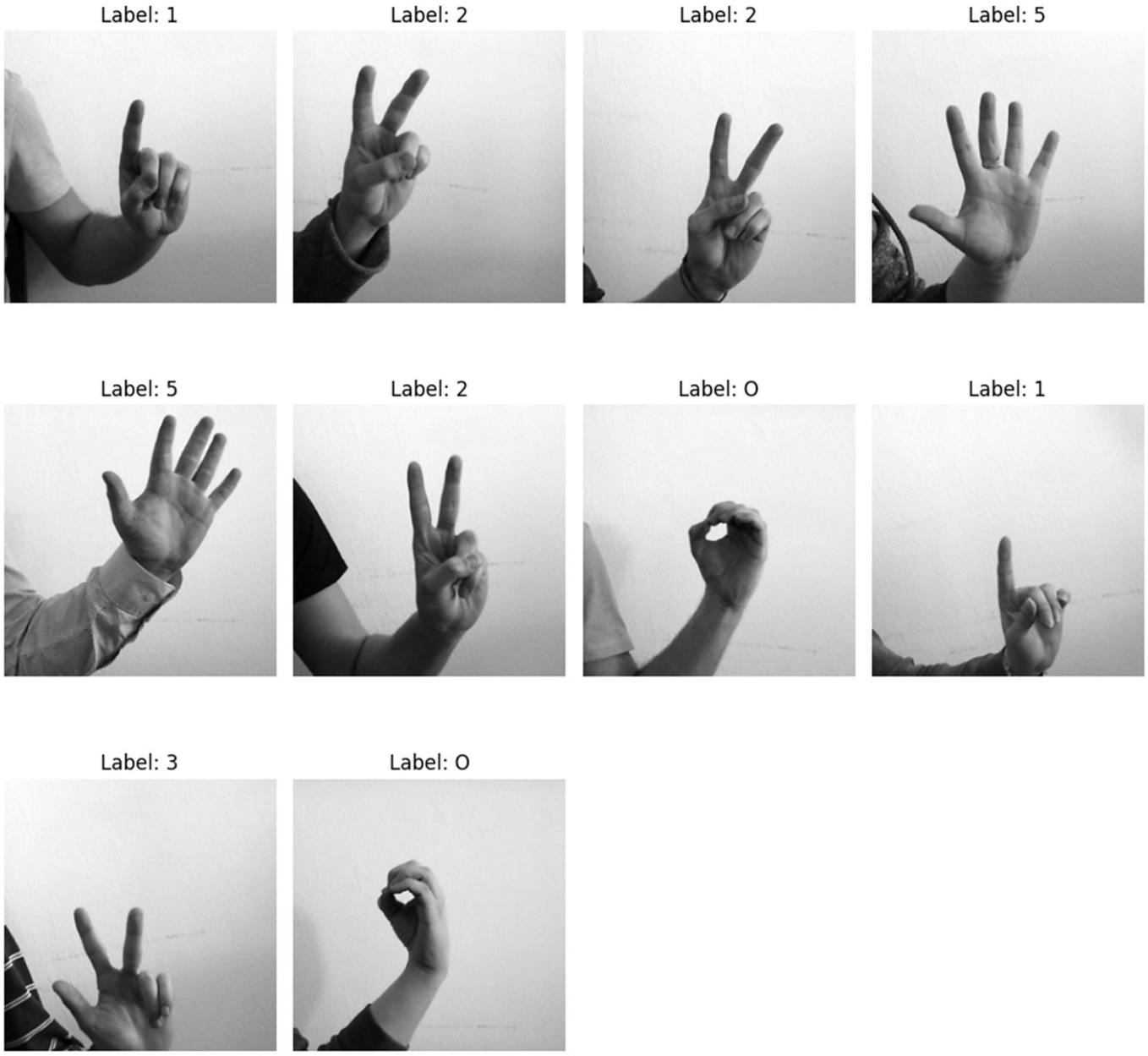


Kontingenčná matica




Predicted: dunno, True: dunno
Predicted: rock, True: rock
Predicted: c, True: c
Predicted: rock, True: rock
Predicted: c, True: c
Predicted: dunno, True: dunno
Predicted: dunno, True: dunno
Predicted: c, True: c

Rozpoznávanie gest ruky z RGB obrazu



Spracovanie dát

- Zmena veľkosti na 220 x 220
- Vloženie do np.array
- Zmena farby na čiernobielu
- One-hot encoding
- Rozdelenie na train 64%, test 20%, val 16%

Vytvorenie CNN siete

- 4 konvolučné + pooling
- 8 plne prepojené

```
def create_custom_serial_model(number_of_classes):
    model = Sequential()

    model.add(Conv2D(32, kernel_size=(3, 3), padding='same', input_shape=(220, 220, 1)))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))

    model.add(Conv2D(64, kernel_size=(3, 3), padding='same'))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))

    model.add(Conv2D(128, kernel_size=(3, 3), padding='same'))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))

    model.add(Conv2D(256, kernel_size=(3, 3), padding='same'))
    model.add(ReLU())
    model.add(BatchNormalization())
    model.add(MaxPooling2D(pool_size=(2, 2)))
    model.add(Flatten())

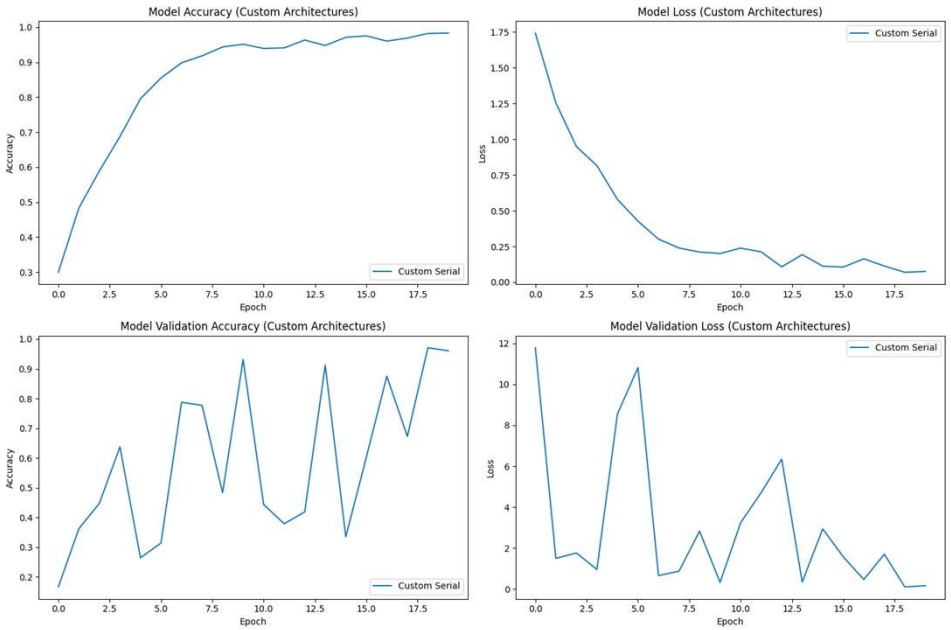
    model.add(Dense(1024))
    model.add(ReLU())
    model.add(Dense(256))
    model.add(ReLU())
    model.add(Dense(128))
    model.add(ReLU())
    model.add(Dense(64))
    model.add(ReLU())
    model.add(Dense(1024))
    model.add(ReLU())
    model.add(Dense(256))
    model.add(ReLU())
    model.add(Dense(128))
    model.add(ReLU())
    model.add(Dense(64))
    model.add(ReLU())
    model.add(Dropout(0.5))

    model.add(Dense(number_of_classes, activation='softmax'))
    model.compile(optimizer=Adam(), loss='categorical_crossentropy', metrics=['accuracy'])

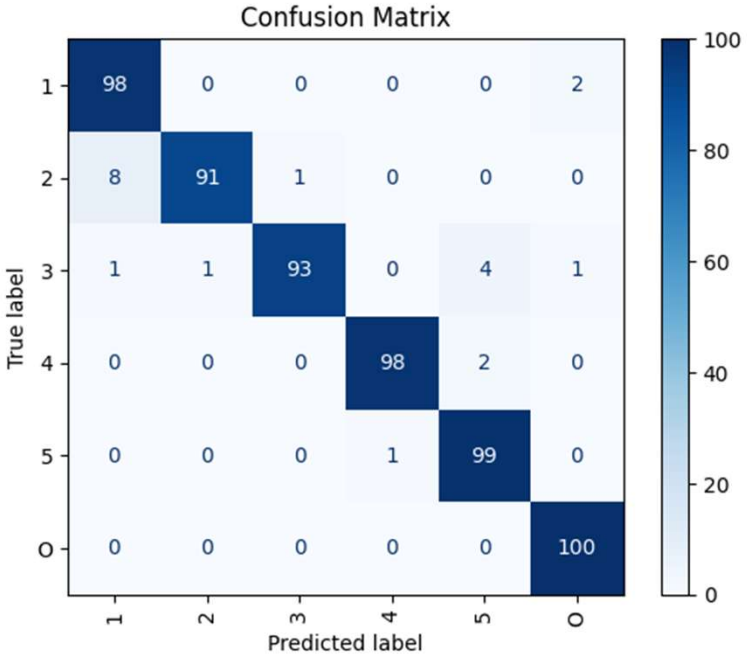
    return model
```

Výsledky trénovania

Proces trénovania



Kontingenčná matica



Predicted: 1, True: 1



Predicted: 5, True: 5



Predicted: 1, True: 1



Predicted: 4, True: 4



Predicted: 4, True: 4



Predicted: 3, True: 3



Predicted: 2, True: 2



Predicted: O, True: O




Ďakujeme za pozornosť